RECEIVER-INDUCED MOMENT EXCHANGE FOR DISTRIBUTED LMB FUSION

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ABSTRACT

Distributed labeled multi-Bernoulli (LMB) trackers can transmit full Gaussian-mixture (GM) messages even when a receiver first collapses each label to one Gaussian. We audit a custom presence-conditioned geometric-average LMB receiver and derive its wire representation from this first irreversible map. Moment matching itself is standard; the contribution is an executable interface certificate spanning label presence, source weighting, deterministic regularization, and serialization. Under the stated admissible numerical contract, sender-side projection preserves every receiver-observable field. A versioned binary codec realizes the interface. In 50 paired eight-sensor trials, moment exchange reduced attempted application-layer payload by 58.28% (95% bootstrap interval: 57.92–58.64%). All 1,119,037 audited retained post-step label instances matched exactly. More generally, the result suggests deriving messages from a receiver's first irreversible map; exactness here remains limited to this receiver and does not extend to mixture-aware or multi-round consumers.

***Index Terms*—** Distributed multi-object tracking, labeled random finite sets, LMB filter, KLA fusion, communication-efficient distributed tracking

1. INTRODUCTION

Distributed multi-object tracking combines sensor views without centralizing measurements. Labeled multi-Bernoulli (LMB) filters represent target existence, state uncertainty, and identity [1, 2, 3]. Their Gaussian-mixture (GM) messages may contain several components per label. The audited software uses a custom presence-conditioned geometric-average receiver, not full-source LMB-density KLA. It serializes every component although this receiver first collapses each label to one Gaussian.

Communication-aware RFS methods alter transmission times, select GM components, or form compact state estimates [4, 5, 6, 7]. Moment-preserving GM reduction is also standard [8]. We propose neither a new projection nor a density-level fusion rule. Instead, we ask which fields can affect a fixed receiver after serialization.

The receiver performs one synchronous round per filtering step. It normalizes fixed degree-based weights over sources carrying each label, projects each GM Bernoulli to existence and two moments, then applies a geometric-average update. Writing the receiver as $\mathcal{F}_{\omega,\mathcal{R}} = \mathcal{G}_{\omega,\mathcal{R}} \circ \mathcal{P}$, we test whether its first irreversible map $\mathcal{P}$ can move across the codec $\mathcal{T}$. The proof must preserve label presence, effective weights, Gaussian overlap, and regularization (Fig. 1).

We contribute an executable receiver contract, a versioned binary codec with adversarial property tests, and a frozen 50-trial paired evaluation. Moment exchange reduced attempted bytes by 58.28%, with zero residual across 1,119,037 retained post-step label comparisons. More broadly, the study illustrates how to derive a message from a consumer's first irreversible map. It claims neither equal mixture densities nor sufficiency for mixture-aware or multi-round use.

2. RELATED WORK

Random finite-set filters model uncertain cardinality and association. PHD filtering gives a first-moment recursion [9]; labeled RFS/LMB models retain identity [1, 2], and multi-sensor variants support cooperation [3]. Under unknown cross-correlations, KLA/GCI forms a normalized geometric mean [10], with PHD and multi-Bernoulli RFS formulations [11, 12]. Consensus and robust LMB/LRFS methods address networked tracking, including label-set inconsistency [13, 14, 15, 16]. Minimum-information- loss and arithmetic-average LMB rules instead use different density-level objectives [17, 18]. These works determine how labeled posteriors are fused. We propose no new density-level rule and audit only the custom presence-conditioned per-label update defined below.

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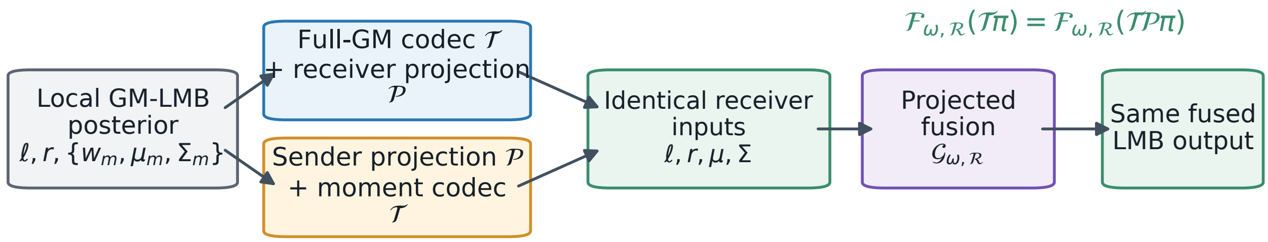

Fig. 1. Receiver-induced message factorization. The full path encodes the GM and projects after decoding; the moment path projects before value-preserving encoding. Under Proposition 1's contract, both paths present identical fields to $\mathcal{G}_{\omega,\mathcal{R}}$ and produce the same LMB output.

Communication methods alter a different layer. Event-triggered LMB methods change when updates are transmitted [4, 5]; related triggers act on $M\delta$-GLMB labels or hypotheses [19]. Partial-consensus PHD changes which GM components are disseminated [6]. State and covariance projection constructs compact track-level inputs for asynchronous Gaussian fusion [7]. In contrast, we hold the fusion rule, schedule, selection, and receiver fixed, then derive the message from that receiver's first projection. The certificate covers label/existence, Gaussian overlap, missing-label normalization, and the codec. Triggering could be composed with this interface, but that combination is not evaluated here.

3. RECEIVER-INDUCED MOMENT EXCHANGE

3.1. Canonical Label-Wise Projection

At time k, source j maintains an LMB posterior

$$\pi_j = \{(r_j^\ell, p_j^\ell) : \ell \in \mathcal{L}_j\},$$
$$p_j^\ell(x) = \sum_{m=1}^{M_j^\ell} \widetilde{w}_{jm}^\ell \mathcal{N}(x; m_{jm}^\ell, P_{jm}^\ell). \tag{1}$$

The implementation maps nonfinite or negative component weights to zero and normalizes the remainder; it uses uniform weights only when all mapped weights vanish. Let $\mathcal{C}(A) = (A + A^\top)/2$. The canonical label-wise projection $\mathcal{P}$ retains (ℓ, r_j^ℓ) and computes

$$\mu_j^\ell = \sum_m \widetilde{w}_{jm}^\ell m_{jm}^\ell, \tag{2}$$

Let $e_{jm}^\ell = m_{jm}^\ell - \mu_j^\ell$. Then

$$\Sigma_j^\ell = \mathcal{C}\left(\sum_m \widetilde{w}_{jm}^\ell \left[\mathcal{C}(P_{jm}^\ell) + e_{jm}^\ell (e_{jm}^\ell)^\top\right]\right). \tag{3}$$

Thus, $\mathcal{P}$ replaces each mixture by the Gaussian with the same first two canonical moments. On its finite numerical domain, it is idempotent: $\mathcal{P}(\mathcal{P}(\pi)) = \mathcal{P}(\pi)$.

3.2. The Executed Presence-Conditioned Receiver

Receiver i aligns the union of source labels and applies $\mathcal{P}$ to each delivered source. Let $\mathcal{S}_i^\ell$ contain self and delivered neighbors carrying label ℓ. The confirmatory path uses fixed symmetric degree-based weights γ_{ij} for both spatial and Bernoulli fusion. On the regular 4+4 graph, $\gamma_{ii} = 1/3$ and each of four neighbors has $\gamma_{ij} = 1/6$. The effective weights are normalized over the participating source subset:

$$\omega_{ij}^\ell = \frac{\gamma_{ij}}{\sum_{q \in \mathcal{S}_i^\ell} \gamma_{iq}}, \qquad j \in \mathcal{S}_i^\ell. \tag{4}$$

Each participating-label update has geometric-average/KLA form. The resulting source-indexed rule is not a full-source LMB-density KLA that treats a missing label as $r^\ell = 0$. The broader software accepts distinct spatial and existence weights, but that generalized receiver is outside the claim and frozen experiment.

The deterministic Cholesky regularizer $\mathcal{R}$ returns $\mathcal{C}(A)$ if factorization succeeds; otherwise it returns the first successful $\mathcal{C}(A) + 10^t \delta I$, where $\delta = 10^{-12} \max(1, \max|\operatorname{diag}(A)|)$ and $t = 0, \dots, 15$. With $S_j^\ell = \mathcal{R}(\Sigma_j^\ell)$, the executed canonical update is

$$K_i^\ell = \mathcal{R}\left(\sum_{j \in \mathcal{S}_i^\ell} \omega_{ij}^\ell (S_j^\ell)^{-1}\right),$$
$$h_i^\ell = \sum_{j \in \mathcal{S}_i^\ell} \omega_{ij}^\ell (S_j^\ell)^{-1} \mu_j^\ell, \tag{5}$$
$$\bar{\mu}_i^\ell = (K_i^\ell)^{-1} h_i^\ell, \qquad \bar{\Sigma}_i^\ell = \mathcal{R}((K_i^\ell)^{-1}).$$

Define $\lambda(A) = d \log(2\pi) + \log|A|$, evaluated from the Cholesky factor. The implementation's overlap factor is

$$\log \eta_{i,\mathcal{R}}^\ell = -\frac{1}{2} \sum_{j \in \mathcal{S}_i^\ell} \omega_{ij}^\ell \left[(\mu_j^\ell)^\top (S_j^\ell)^{-1} \mu_j^\ell + \lambda(S_j^\ell)\right]$$
$$+ \frac{1}{2}\left[(\bar{\mu}_i^\ell)^\top K_i^\ell \bar{\mu}_i^\ell + \lambda(\bar{\Sigma}_i^\ell)\right]. \tag{6}$$

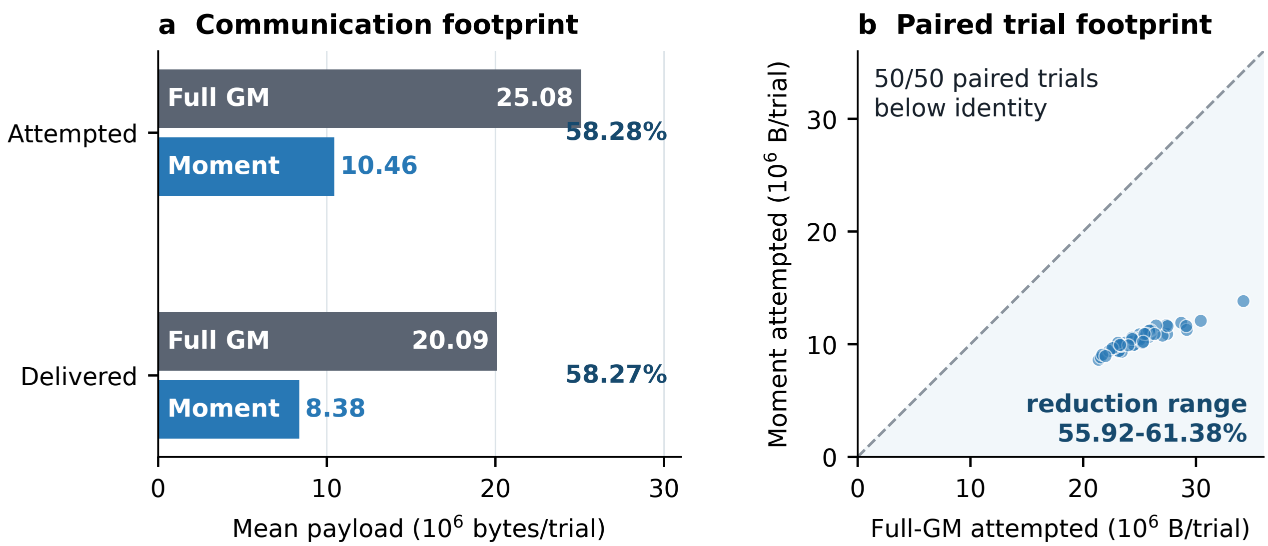

Fig. 2. Communication evidence from 50 paired trials. Left: mean attempted and delivered application-layer payloads; percentages are mean per-trial reductions. Right: each point is one paired trial; all lie below the no-saving identity. The retained-field audit is reported in Sec. 4.

Without added jitter, Eq. (6) is the weighted Gaussian- product integral. With jitter, it exactly reproduces the implemented regularized scalar rather than the unregularized analytic integral. With products below over $j \in \mathcal{S}_i^\ell$, the Bernoulli update is

$$q_1^\ell = \eta_{i,\mathcal{R}}^\ell \prod_j (r_j^\ell)^{\omega_{ij}^\ell}, \; q_0^\ell = \prod_j (1 - r_j^\ell)^{\omega_{ij}^\ell}, \; \bar{r}_i^\ell = \frac{q_1^\ell}{q_0^\ell + q_1^\ell}. \tag{7}$$

Each r_j^ℓ is clamped to $[10^{-9}, 1 - 10^{-9}]$, and the final denominator is lower-bounded by machine epsilon. Denote Eqs. (5)–(7) by $\mathcal{G}_{\omega,\mathcal{R}}$ and the receiver by $\mathcal{F}_{\omega,\mathcal{R}} = \mathcal{G}_{\omega,\mathcal{R}} \circ \mathcal{P}$.

Let $\mathcal{T}$ be the posterior-content map of the versioned encode/decode transport below. It preserves labels, r, weights, and means, and replaces each component covariance by $\mathcal{C}(P)$. A message is *admissible* when $\mathcal{P}$ returns finite fields, every call to $\mathcal{R}$ succeeds, and all quadratic, log-determinant, exponential, and Bernoulli intermediates remain finite. On this numerical domain, $\mathcal{P} \circ \mathcal{T} = \mathcal{P}$ and $\mathcal{T} \circ \mathcal{P} = \mathcal{P}$.

Proposition 1 (executed fusion-output equivalence). Let $\boldsymbol{\pi}_{-i}$ be the delivered-neighbor tuple at receiver i. Assume admissible inputs, identical active-label presence, delivery masks, and effective source weights after presence normalization; common $\mathcal{C}$, $\mathcal{P}$, $\mathcal{R}$, and $\mathcal{T}$; and disabled mode-aware weighting and covariance inflation. Then

$$\mathcal{F}_{\omega,\mathcal{R}}(\pi_i, \mathcal{T}\boldsymbol{\pi}_{-i}) = \mathcal{F}_{\omega,\mathcal{R}}(\pi_i, \mathcal{T}\mathcal{P}\boldsymbol{\pi}_{-i}). \tag{8}$$

Proof. The identities above give $\mathcal{P}\mathcal{T}\pi_j = \mathcal{P}\mathcal{T}\mathcal{P}\pi_j = \mathcal{P}\pi_j$. They preserve each source-label's r, μ, Σ and presence in $\mathcal{S}_i^\ell$. Hence the active ω_{ij}^ℓ, regularized S_j^ℓ, canonical K_i^ℓ, h_i^ℓ, overlap $\eta_{i,\mathcal{R}}^\ell$, and q_0^ℓ, q_1^ℓ are unchanged, yielding identical $\bar{r}_i^\ell, \bar{\mu}_i^\ell, \bar{\Sigma}_i^\ell$. $\square$

Define receiver equivalence by $\pi \sim_{\mathcal{P}} \pi'$ when $\mathcal{P}\pi = \mathcal{P}\pi'$. We call $\mathcal{P}(\pi)$ *fusion-sufficient* only for this receiver, for which $\mathcal{F}$ is constant on each such class. Equation (8) does not imply equal posterior densities. It can fail for a mixture-aware consumer, different label selection or numerical maps, quantization, covariance inflation, or multi-round reuse.

3.3. Versioned Message and Byte Accounting

Both arms use the same versioned, little-endian binary codec underlying $\mathcal{T}$: a 24-byte message header, a 20-byte per-label object-record header, and binary64 component fields for weight, mean, and the upper covariance triangle. The encoder applies $\mathcal{C}$ before packing, and the decoder reconstructs the symmetric binary64 covariance. For state dimension d, a message with M_ℓ components per label has

$$B = 24 + \sum_\ell \left[20 + 8M_\ell \left(1 + d + \frac{d(d+1)}{2} \right) \right] \text{ bytes.} \tag{9}$$

The full message retains every M_ℓ in Eq. (1); the proposed message sets $M_\ell = 1$ using Eqs. (2)–(3). Attempted bytes are counted after encoding and before the drop draw; delivered bytes count messages decoded after successful delivery. No tracker variable, scheduling decision, or fusion parameter changes.

4. EXPERIMENTS

4.1. Paired Confirmatory Protocol

We evaluate an eight-sensor 4+4 formation for 100 steps. Each node runs the same LMB recursion and one synchronous fusion round per step. The 4-D constant- velocity model uses 2-D position measurements and ten targets born eight steps apart; moving sensors have a $60°$ half-angle field of view. The fixed graph has two four-node cliques and four cross links. Retained labels satisfy $r > 10^{-2}$. Detection probability is 0.9, mean clutter is 3, and measurement-noise standard deviation is 3. Per-node drop probabilities $[0, 0.1, 0.2, 0.5]$ occur with multiplicities $[1, 4, 1, 2]$ in random trial-wise assignments (mean 0.2).

We compare only periodic full GM fusion messages and fusion-sufficient moment messages. The 50 paired trials use seeds 82–131. Within each pair, target trajectories, measurements, schedules, labels, fixed degree-based weights, and drop uniforms are identical; attempted and delivered masks are checked for equality. Local accuracy uses Euclidean OSPA (cutoff $c = 5$, order $p = 2$), while agreement uses mean pairwise OSPA, position disagreement, and cardinality dispersion [20]. The primary statistic is the mean of 50 per-trial attempted-byte reductions. Its paired percentile-bootstrap interval uses 10,000 resamples and fixed seed 20270710.

Independent property tests recompute mixture moments and exercise asymmetric, ill-conditioned, and singular covariances. Codec tests round-trip canonical fields, while mutation tests reject perturbed labels, r, μ, or Σ. A negative control confirms that enabling covariance inflation breaks the claimed equivalence.

Table 1. Confirmatory means ($N = 50$). Att./Del.: attempted/delivered application payload in 10^6 bytes per trial; Local/Cons.: OSPA to truth/mean pairwise sensor OSPA ($c = 5, p = 2$); lower is better.

Message	Att.	Del.	Local	Cons.
Full GM	25.084	20.091	2.076	1.951
Moment msg.	10.458	8.378	2.076	1.951

4.2. Results and Scope

Table 1 and Fig. 2 show a 58.277% mean per-trial reduction in attempted bytes (95% interval: 57.923–58.636%; minimum 55.922%). The mean delivered-byte reduction is 58.267%.

The field audit is stronger than rounded metric agreement. Snapshots are taken after the common $r > 10^{-2}$ retention step. All 40,000 paired post-step sensor-time snapshots have identical label sets. Across 1,119,037 retained label-instance comparisons, the maximum residuals in existence probability, mean, and covariance are each exactly zero. Consequently, local E-OSPA, consensus OSPA, position disagreement (4.614), and cardinality dispersion (0.203) are identical in every paired trial. This matches the filter-level consequence of Proposition 1 and checks the serialization boundary; raw pre-retention equality is established by the proposition.

5. DISCUSSION

Proposition 1 has short algebra but a substantial implementation contract. The full arm transports every GM component before $\mathcal{P}$; the moment arm projects before transport. The identities $\mathcal{P}\mathcal{T} = \mathcal{P}$ and $\mathcal{T}\mathcal{P} = \mathcal{P}$ include covariance canonicalization. Both arms then share label alignment, effective weights, $\mathcal{R}$, and Bernoulli fusion.

With identical initial conditions, measurements, and masks, deterministic retention, prediction, and update extend the single-round equality through the frozen recursion.

Equation (9) makes the 58.28% reduction workload specific. For $c_d = 1+d+d(d+1)/2$, fixed labels save $8\sum_\ell(M_\ell - 1)c_d$ bytes. Savings vanish at $M_\ell = 1$ and grow with dimension and mixture multiplicity. The bootstrap interval captures component-count variability in this workload, not cross-model generality.

Equivalence can fail under mixture-dependent gating or weights, different pruning, quantization, covariance inflation, or later mode reuse. Exactness also requires common $\mathcal{C}$, $\mathcal{P}$, $\mathcal{R}$, $\mathcal{T}$, and binary64 conventions. Distinct spatial and existence weights in the broader software are outside this certificate.

Counts include application headers but exclude link framing, retransmission, latency, and energy. This is not entropy coding or rate optimality; hardware claims require target-packet measurements.

The reusable procedure is to identify a consumer's first irreversible map, certify downstream factorization, promote its output to a message, and test after serialization. Here evidence remains limited to one eight-sensor model, graph, binary64 codec, and synchronous round.

6. CONCLUSION

We derived a receiver-induced moment message for a custom presence-conditioned geometric-average LMB receiver. Sender-side projection preserved every audited retained post-step field and reduced attempted application-layer payload by 58.28% across 50 paired trials. The result supports a broader interface-design principle: derive a message from the receiver's first irreversible map and certify the downstream numerical contract. Exactness here remains specific to this receiver and does not extend to mixture-aware or multi-round consumers.

COMPLIANCE WITH ETHICAL STANDARDS

This is a numerical simulation study for which no ethical approval was required.

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